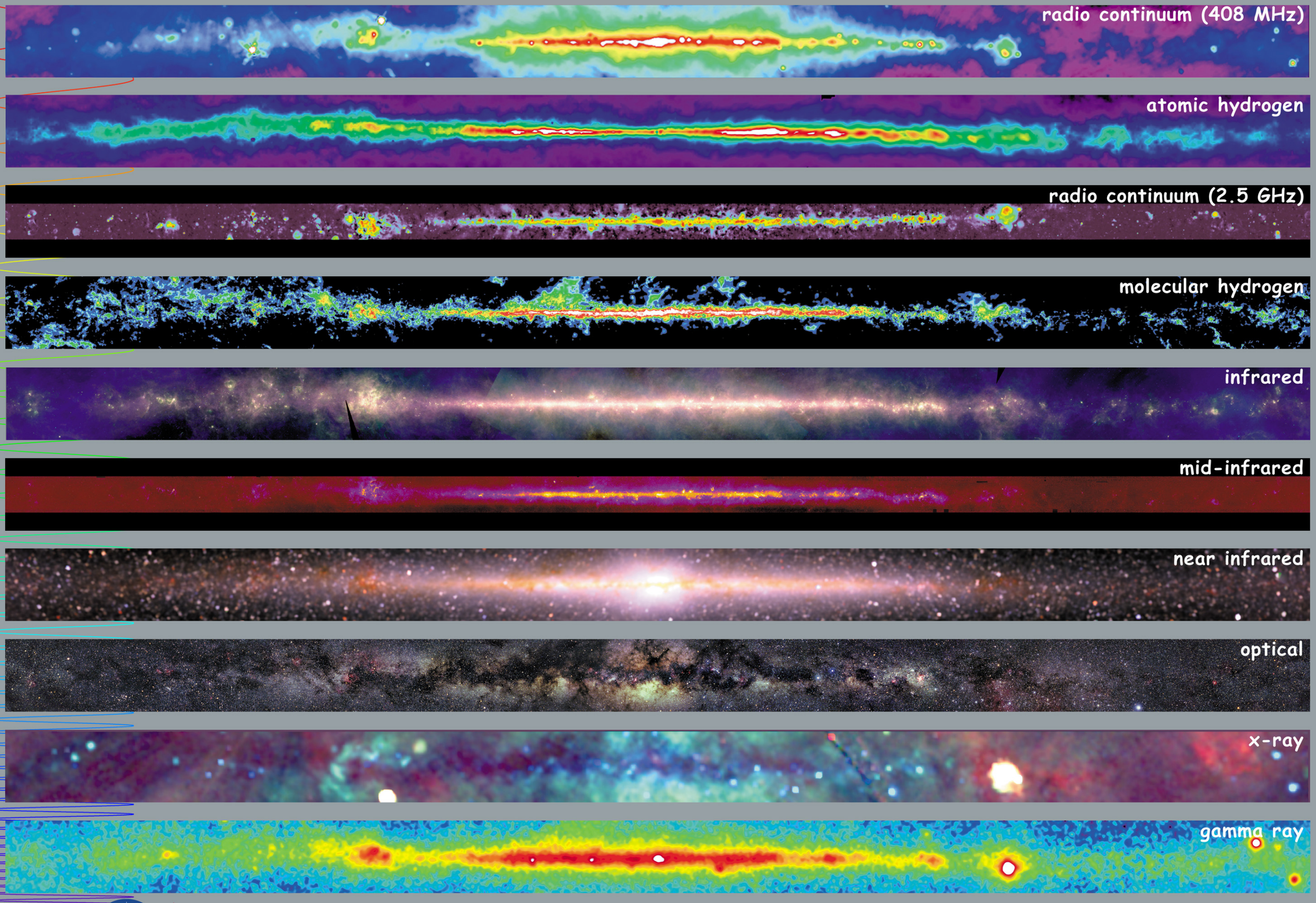
Observational Astronomy

- Overview
- Coordinates on the Sky
- Radiation
  - Basic definitions
- The Atmosphere
  - Absorption
  - Scattering
  - Emission
  - Turbulence and Seeing
- Telescopes
  - Designs
  - Field of View
  - Adaptive Optics
- Imaging
  - Filters
  - Detector
    - CCDs
      - QE
      - Noise -Dark, Readout, Photon
      - Defects
      - Cosmic Rays
      - Calibration
      - Drift scanning
- Photometry and S/N
- Spectroscopy
  - Dispersers
  - Resolution
  - Slits vs. fibres vs. Integral Fields
- Interferometry
- Radio
- Interstellar Extinction
- Space Astronomy?

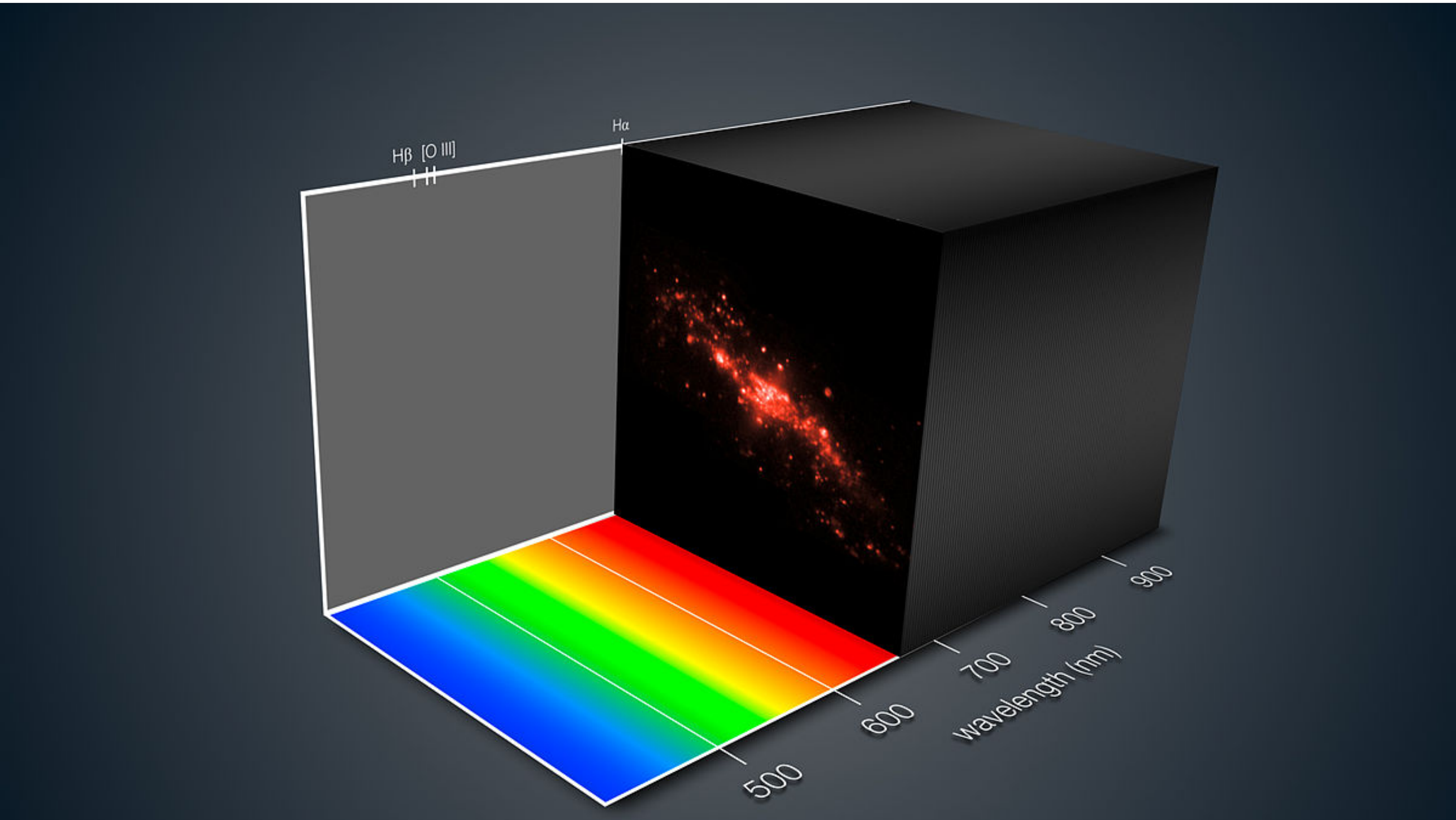
- Databases



Multiwavelength Milky Way

http://adc.gsfc.nasa.gov/mw

Universe as a Data Cube



By ESO/MUSE consortium/R. Bacon/L. Calçada - http://www.eso.org/public/images/eso1407a/, CC BY 4.0, https://commons.wikimedia.org/w/index.php?curid=31545397

Universe as a Data *Sphere*

- Two angular coordinates, one spectral.
- Angular coordinates:

$$4\pi \times (180/\pi)^2 = 41{,}252\,\mathrm{deg}^2 = 5.34 \times 10^{11}\,\square''$$

- At Hubble resolution 0.1” there are 100 elements per square arcsec
- 10^{14} resolution elements in the sky

Universe as a Data *Sphere*

- Spectral coordinates:
  - Astronomical spectra range from 50 Mhz radio to 10 TeV gamma rays, or ?? Hz
    - 21 orders of magnitude or 50 e-folds.
- High spectral resolution (in optical) has

$$R = \frac{\lambda}{\Delta\lambda} \sim 100{,}000 \qquad\qquad d\ln\lambda = \frac{\Delta\lambda}{\lambda} \sim 10^{-5}$$

  - Taking uniform steps in of gives ~10^7 resolution elements spectrally

Universe as a Data *Sphere*

- Total number of data points could be about ~10^{21}
- Of course this is for the *static* sky
- …and ignores polarization

Universe as a Data *Sphere*

- Total number of data points could be ~10^{21}

  - Allowing for storing the signal and the noise digitally adds another factor of ~10, so 10 zettabytes.

  - Large data centres contain exabytes, so within reach with help from Moore's Law

  - Some data compression may be possible since much of the sky is dark.

But the problem is not *storage ... its collecting the data!*

Universe as a Data *Sphere*

- Total number of data points could be ~10^{21}
- Actual observations are not efficient enough to measure this in our lifetime:
    - A few hours on an 8m class telescope will get a redshift from the brightest emission line of a faint z=1 galaxy.
        - One can get perhaps 100 of these with individual slits of a few square arcseconds (say ~1000 resolution elements on the sky)
        - In the spectral dimension perhaps a few thousand measurements
        - Total number of data points: ~10^{6} !
- And that's in the optical where you can work from the ground ....

The point of this …

- Only a tiny fraction of the information content of the Universe is observed or observable.

- To expand the range of data, need new capabilities

- (But there are always engineering tradeoffs: you can’t have it all)

Current Directions in Imaging

- Area coverage
  - UNIONS, HSC, LSST
- Narrow band filters
- Time Domain
  - LSST
- Angular resolution
  - Space:
    - Euclid, Roman
- Adaptive Optics:
  - All major ground based telescopes.
    - But fields-of-view are still very small

New Directions in Spectroscopy

- Depth:
  - ELT/TMT/GMT, JWST
- Multiplex:
  - SDSS, DESI, PFS, MSE
    - all fibre based
- Wavelength range
  - Ground based:
    - mostly optical/NIR/ submm plus radio
    - space for other wavelengths
  - Spatially resolved spectroscopy
    - IFU, FTS, grism, etc
  - Improving spectral resolution
    - Mostly for stars

- Even with *existing* facilities and instruments there is opportunity to explore new parts of parameter space

- Of course all of this needs to be motivated by a scientific question ….